

NAICS and SIC Industry-Code Coverage in the NPDES Major-Individual Panel, by State (2005–2025)

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Scope and data

This brief measures how often EPA NPDES permits carry an industry classification code—North American Industry Classification System (**NAICS**) and the older Standard Industrial Classification (**SIC**)—and how that coverage varies across states. The universe is the **major-individual** permit population (individual “NPD” permits flagged *major* in a given year), reconstructed by year over 2005–2025 with entry and exit allowed. The unit of observation is a *permit-year*; a permit-year “has NAICS” (“has SIC”) if that permit appears at least once in `NPDES_NAICS.csv` (`NPDES_SICS.csv`). Coverage figures pool all 139,992 permit-years across the 56 reporting states and territories.

Two properties of the source matter. First, NAICS/SIC presence is a *time-invariant permit attribute*: the code files carry no date, so within a permit, coverage does not change year to year—the cross-state variation below is structural, not a reporting trend. Second, “coverage” means a code is *on file*; it does not assess whether the code is correct.

Headline findings

- **SIC is near-universal: 98.7% nationally**, at or near 100% in almost every state. Only ND, VT fall below 90%.
- **NAICS is sparse and sharply state-dependent: 35.9% nationally**. The distribution is bimodal, not centered on the mean: 16 states and territories record NAICS on essentially *none* of their major-individual permits, while 10 record it on 95%+.
- **The gap is administrative, not industrial**. Whether a permit has a NAICS code tracks the *permitting authority*, not the facility. Several of the largest state programs sit at $\approx 0\%$ (e.g. CA, IL, OH, WI, MD, WV; TX at 0.2%), while neighbors of similar industrial mix (e.g. KY, OK, ID at 99–100%) are fully coded.

General state reporting of NAICS and SIC codes

Industry codes enter ICIS-NPDES at permit issuance and are maintained by the administering authority—the state environmental agency in delegated (authorized) states, or the EPA Region otherwise. Because SIC long predates NAICS and was the required classification when most of these permits and their legacy records were created, virtually every major-individual permit carries a SIC code: coverage is 98.7% pooled and effectively saturated in all but a couple of jurisdictions. NAICS adoption, by contrast, was never uniformly back-filled. A state’s ICIS-NPDES records either systematically include NAICS (yielding near-100% coverage) or they do not (yielding near-0%), producing the two-cluster pattern in Table 1 rather than a smooth gradient. The modest national upward drift over the window—from 33.6% (2005) to 38.9% (2025) in Table 2, against flat SIC—reflects newer permits in already-coded states, not laggard states beginning to report.

Implication for analysis. For any cross-state industry classification, use **SIC**: its coverage is uniform, so it does not confound industry with state. Pooled NAICS is unusable, and conditioning on NAICS presence would silently select on state (dropping entire programs such as CA, OH, and IL). This is recorded in `docs/data.quirks.md` as the operative guidance for the panel.

Caveats

- Figures are *permit-year* shares pooled over the window; a state’s permit count is its total active major-individual permit-years, not distinct permits. Because coverage is time-invariant within a permit, pooling does not distort the shares.
- Coverage measures *presence* of a code, not its accuracy or its match to the facility’s true activity.
- “States” includes DC and U.S. territories (56 units); the smallest have very few permit-years, so their percentages are noisy.
- ND, VT are the only sub-90% SIC jurisdictions and warrant a look before use—atypical against the otherwise saturated SIC pattern.

Reproducibility

Coverage is computed by `scripts/diagnostics/naics_sic_coverage_by_state_year.R` (writes a state $\times$ year CSV to `output/tables/`). This brief’s tables are generated from the newest such CSV (`naics_sic_coverage_by_state_year_2026-07-13_1219.csv`) by `scripts/diagnostics/make_naics_sic_coverage_brief.R`, which emits the `\input` fragments and the in-text numbers. Rerun that generator after any refresh; do not edit the fragments or this PDF by hand.

Table 1: NAICS and SIC coverage of major-individual NPDES permits, pooled 2005–2025, by state/territory. n = permit-years with a code on file; % = share of that state’s permit-years.

State	Permit-yrs	NAICS n	NAICS %	SIC n	SIC %
AK	887	873	98.4	866	97.6
AL	3,846	2,809	73.0	3,835	99.7
AR	2,440	2,370	97.1	2,440	100.0
AS	81	0	0.0	81	100.0
AZ	1,495	67	4.5	1,368	91.5
CA	4,909	0	0.0	4,909	100.0
CO	2,608	63	2.4	2,608	100.0
CT	1,904	0	0.0	1,865	98.0
DC	84	0	0.0	84	100.0
DE	328	0	0.0	328	100.0
FL	6,759	6,532	96.6	6,759	100.0
GA	4,175	3,937	94.3	4,101	98.2
GU	97	0	0.0	97	100.0
HI	393	0	0.0	373	94.9
IA	2,761	2,714	98.3	2,714	98.3
ID	820	820	100.0	820	100.0
IL	5,685	0	0.0	5,685	100.0
IN	4,038	1,621	40.1	4,038	100.0
KS	1,065	980	92.0	1,058	99.3
KY	2,874	2,864	99.7	2,874	100.0
LA	4,965	24	0.5	4,965	100.0
MA	2,420	63	2.6	2,420	100.0
MD	1,853	0	0.0	1,853	100.0
ME	1,546	396	25.6	1,546	100.0
MI	3,834	3,490	91.0	3,834	100.0
MN	2,076	0	0.0	1,996	96.1
MO	3,677	670	18.2	3,677	100.0
MP	42	0	0.0	42	100.0
MS	1,894	1,752	92.5	1,894	100.0
MT	709	23	3.2	706	99.6
NC	4,542	21	0.5	4,487	98.8
ND	500	385	77.0	406	81.2
NE	1,045	65	6.2	1,045	100.0
NH	1,010	189	18.7	1,010	100.0
NJ	2,379	621	26.1	2,375	99.8
NM	717	84	11.7	717	100.0
NV	298	144	48.3	298	100.0
NY	6,736	1,376	20.4	6,736	100.0
OH	6,011	0	0.0	6,011	100.0
OK	2,199	2,196	99.9	2,199	100.0
OR	1,518	21	1.4	1,427	94.0
PA	8,072	1,417	17.6	7,796	96.6
PR	1,433	0	0.0	1,433	100.0
RI	499	399	80.0	499	100.0
SC	3,289	3,077	93.6	3,224	98.0
SD	558	548	98.2	558	100.0
TN	3,327	3,150	94.7	3,327	100.0
TX	14,739	35	0.2	14,713	99.8
UT	846	841	99.4	841	99.4
VA	3,043	2,706	88.9	3,043	100.0
VI	121	0	0.0	121	100.0
VT	650	629	96.8	0	0.0
WA	1,424	283	19.9	1,313	92.2
WI	2,468	0	0.0	2,468	100.0
WV	1,905	0	0.0	1,905	100.0
WY	398	15	3.8	382	96.0
All	139,992	50,270	35.9	138,170	98.7

Table 2: National coverage by year (pooled across states).

Year	Permit-yrs	NAICS %	SIC %
2005	5,864	33.6	99.1
2006	6,180	33.7	98.9
2007	6,422	33.4	98.9
2008	6,554	33.7	98.8
2009	6,650	33.6	98.8
2010	6,713	33.7	98.7
2011	6,810	34.5	98.5
2012	6,822	34.7	98.5
2013	6,851	35.1	98.5
2014	6,839	35.3	98.5
2015	6,819	35.6	98.6
2016	6,880	36.5	98.6
2017	6,892	36.9	98.6
2018	6,870	37.2	98.6
2019	6,826	37.3	98.6
2020	6,795	37.6	98.7
2021	6,773	37.8	98.7
2022	6,750	37.9	98.7
2023	6,676	37.9	98.7
2024	6,544	38.6	98.8
2025	6,462	38.9	98.8